Process Dynamics and Control

16 lectures given to CEBT II by Dr Sebastian Ahnert (sea31) in Michaelmas Term 2026.
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1. Introduction

1.1 Introduction

The world is a dynamic place in which steady state is seldom reached: there is time dependence. All physicochemical processes employed by chemical engineers are similar in that they are constantly responding to changes. Changes may be, for example, fluctuations in feed flow rate, temperature or composition, or they may be imposed by the operator to deal with a new product specification. The response to a change is dynamic behaviour, which necessitates a means of control to ensure that process parameters are maintained within desired limits, or specifications are met. The requirements of control systems have become much more demanding in recent years because of increased competition, stricter environmental and safety regulations, and also because of the increased complexity and integration of modern plants. In large plants, such as oil refineries, thousands of variables will be monitored and controlled.

Chemical engineers must be able to describe the dynamics of a process and then, based on an understanding of its behaviour, design the appropriate control system. The design of a process plant must consider from the outset how the plant is to be controlled, as state-of-the-art control can never entirely change the effect of decisions introduced by process design: however good a driver you are, you can't make a family saloon perform like a racing car.

The broad aims of this course are to enable you to:

- Analyse the dynamic behaviour of processes.
- Design and analyse controllers and control systems for stability and control of quality.
- Devise advanced control systems.
- Devise control strategies for unit operations

The course contains four main sections:

Section 2 – Dynamics of Linear Systems

We will review and extend the Process Dynamics part of CEBT IB Engineering Mathematics.

Section 3 – Feedback Controller Design

We will study feedback control of a single input/single output system, including proportional, integral and derivative control, frequency response, stability and controller tuning.

Section 4 – Advanced Control Systems

We will study aspects of advanced control systems, including cascade, feedforward and ratio control, and the control of nonlinear systems.

Section 5 – Process Control Systems Design

We will consider multivariable control systems, and comment on heuristics and guidelines for the design of control systems for whole plants.

Books

Marlin, Thomas E., *Process Control: Designing Processes and Control Systems for Dynamic Performance*, McGraw Hill, 2000. This book has been made available by the author at http://pc-textbook.mcmaster.ca/ (accessed 2/8/2019).

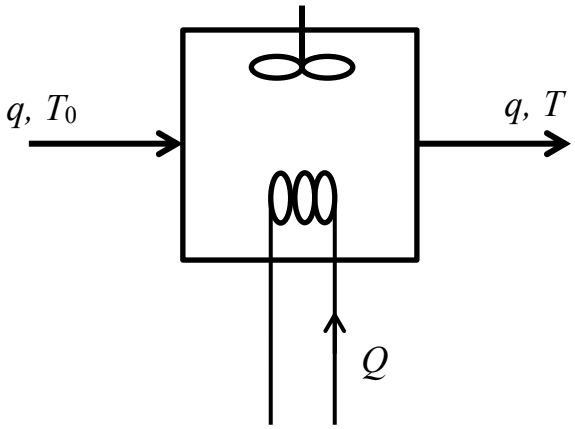
Seborg, Dale E., Edgar, Thomas F., Mellichamp, Duncan A. and Doyle, Francis J. III, *Process Dynamics and Control*, Wiley, 3rd Edition, 2011.

Stephanopoulos, George, *Chemical Process Control: An Introduction to Theory and Practice*, Prentice-Hall, 1984.

1.2 Introduction to Control Issues

Example 1.1: Heating a flow of liquid in a stirred tank

Liquid of density ρ flows at steady volumetric flow rate q through a well-stirred tank of constant volume V. The inlet temperature is T_0 and the temperature of the liquid in the tank is T. Heat is supplied at rate Q.



At steady state design conditions

$$\rho q C_p T_0 + Q = \rho q C_p T$$

We will be interested in how to deal with changes of two sorts:

1. How to adjust Q to maintain T at the desired value – its set point T_{sp} – if there is a disturbance to the system, for example a change in inlet temperature T_0. This is known as regulator control.

2. How to adjust Q to change T to a new desired value of T_{sp}, for example to satisfy a change in product specification. This is known as servo control.

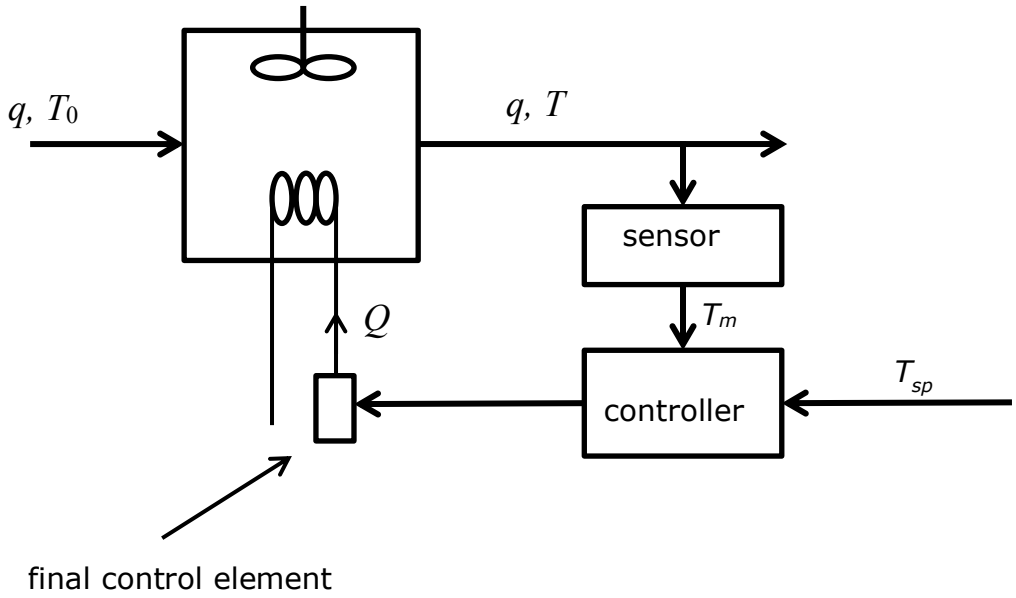
Here T is the controlled variable, Q is the manipulated variable and T_0 is the disturbance variable. Note that there may be other controlled variables, e.g. the volume of the tank V, other disturbance

variables, e.g. the flow rate q, and other manipulated variables, e.g. the flow rate q – we must specify what is what. Note too that what we can do depends on what we can measure.

These operations may occur over different timescales. They have somewhat different objectives: for regulator control the aim is to minimise the effect of a disturbance on a specified process variable, whereas for servo control the aim is to impose a change on the system, because of a new specification for a process variable.

1.3 Feedback Control

Referring to Example 1.1, we could measure the temperature in the tank, T_m, compare that to the set point temperature, and adjust Q to make T_m approach T_{sp}.



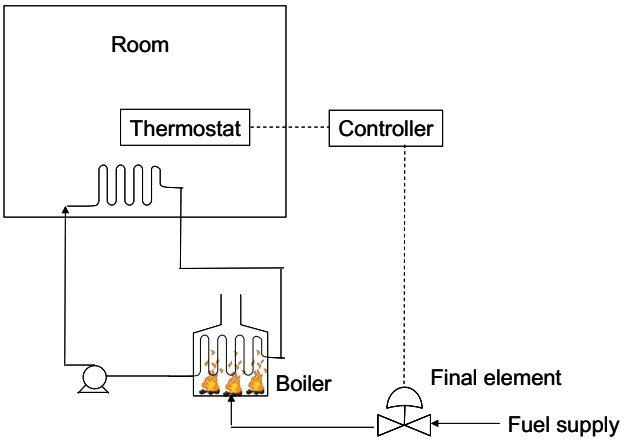
The value of the error ϵ

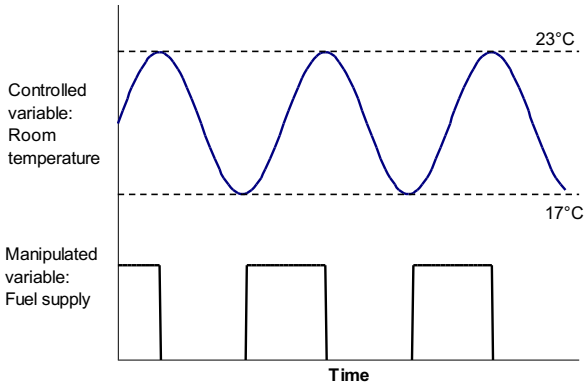
$$\epsilon = T_{sp} - T_m$$

is fed back, and Q is adjusted accordingly, by the final control element, in this case a heater. This is feedback control, and Section 3 of this course is devoted to it. Note that in order for any control action to be made, an error, as defined above, must occur. An important feature of the *feedback loop* is that it can introduce instability to an otherwise stable process.

Example 1.2: Central heating system

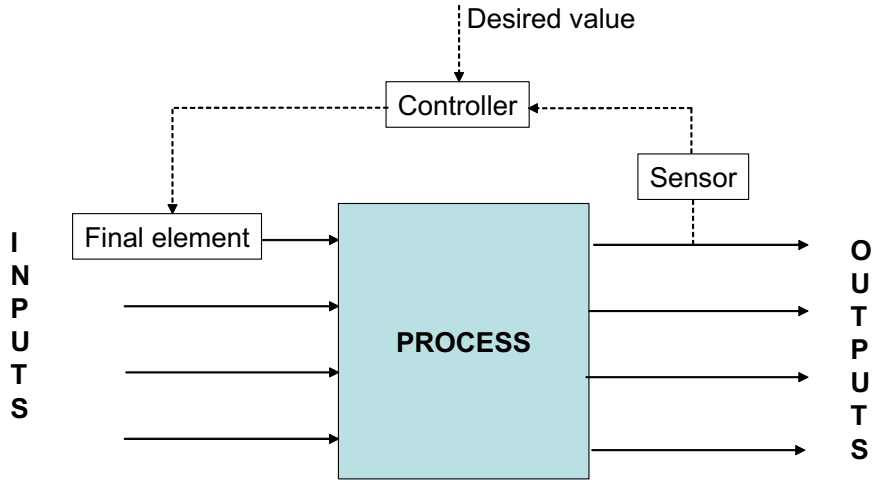
A familiar example is the home central heating system. The room temperature is measured and compared to the desired setting on the thermostat. If the temperature is higher than desired, then the boiler will be turned off, or if the temperature is lower than desired, the boiler will be turned on.





This gives crude control, with the room temperature varying over a range, even when the outside temperature is constant. We will generally want to do better than this on a process plant.

A generalised model for process control is as follows.

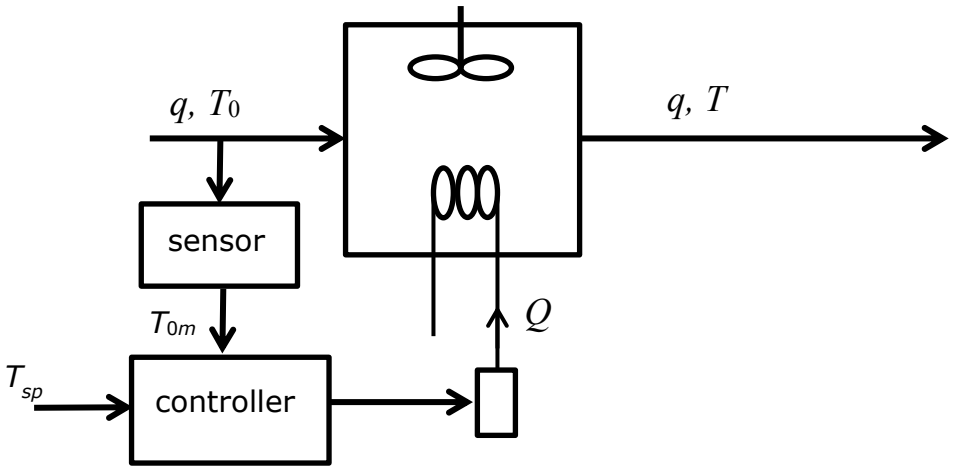


A feedback controller measures the deviation of an output variable from its desired value (set point) and adjusts an input variable to make the correction. The control *loop* consists of the sensor, controller and final element (often a control valve).

We will find that the use of feedback control with a stable process can lead to oscillations, or even instability, in the output.

1.4 Feedforward Control

A different approach to deal with disturbances in the inlet temperature T_0, for example, is to measure it, and adjust Q accordingly.



T_0 is measured, and the information is fed forward to a model of the process. The model is used to calculate the value of Q which will maintain T at its set point. In contrast to feedback control, feedforward control allows control action to occur without there being an error in the controlled variable. But for perfect control a perfect model is needed. Feedforward control, and other advanced control techniques, will be discussed in Section 4.

1.5 Control Objectives

Control objectives fall into three main categories all of which need to be considered when designing the control system for a process plant:

Safety – Control systems are one of the lines of defence employed to prevent safety and environmental constraints being breached but they are **not solely** relied upon as they can fail. Hence additional layers of higher integrity equipment incorporating shutdown logic with additional sensors and final elements are also incorporated in the design with a mechanical relief valve being the ultimate safety system.

Operability – This term is used to describe the notion of being able to make the plant do what the operator wants rather than leaving it to its natural inclinations. The majority of control loops are associated with operability and this objective is achieved in the design of the individual control loops. Specific flow rates, temperatures, pressures and levels need to be set and maintained in order for the plant to perform.

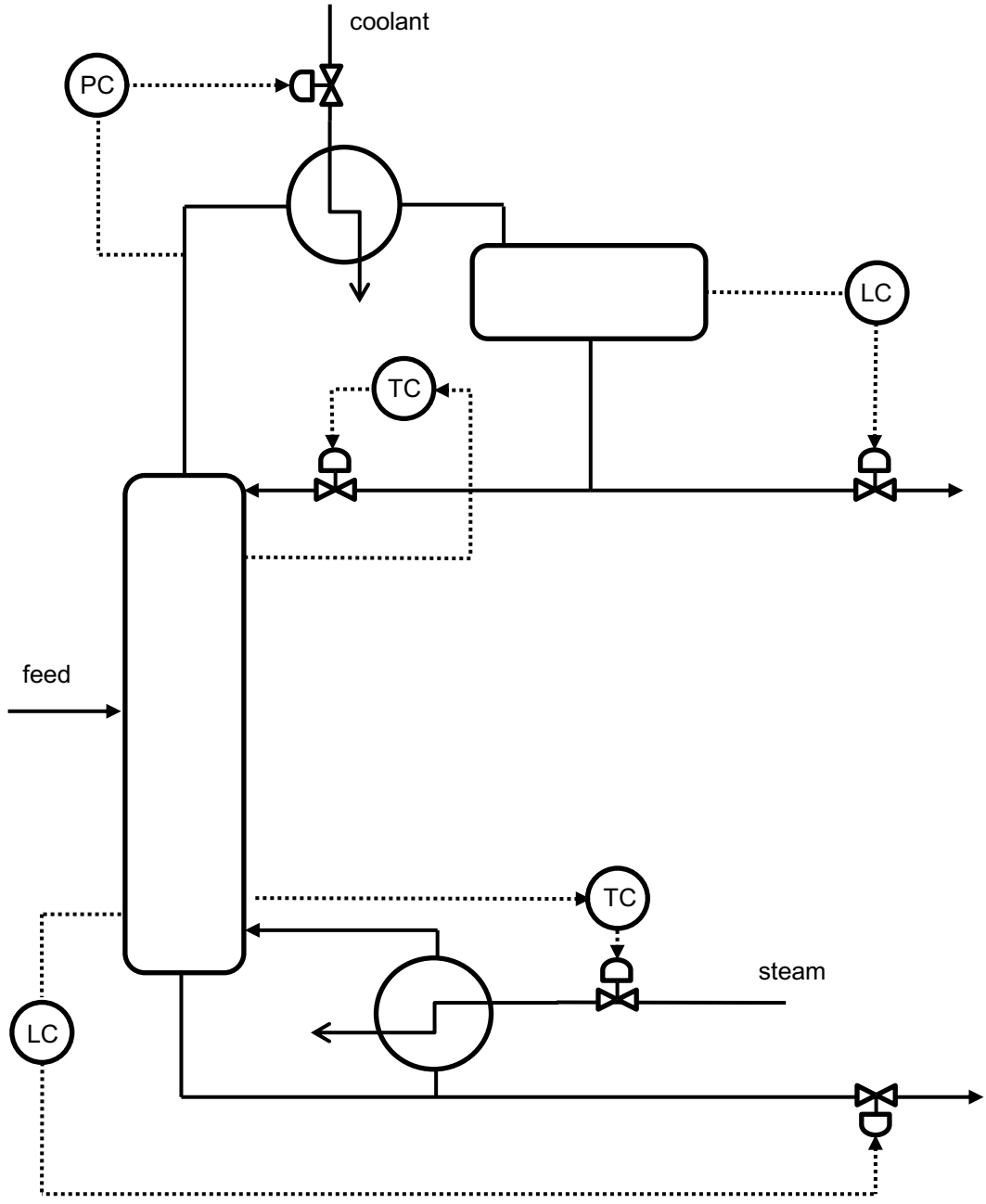
Profitability - The top level of process control is the *strategic* control level and is concerned with achieving the desired production rate, product quality and energy efficiency. This objective is achieved by placing the individual control loops in the correct part of the flowsheet and ensuring that there are no conflicts between individual controllers. It can also often involve the use of *advanced* control systems that govern how control loops operate together.

This course is primarily focused on the operability and profitability objectives.

1.6 Distillation column control

Example 1.3: Distillation control system

A typical two component distillation uses 5 control loops to control the column and achieve the top and bottom product quality. A possible control structure is shown below.



If the feed is the product of an upstream unit, it may not be possible to control it.

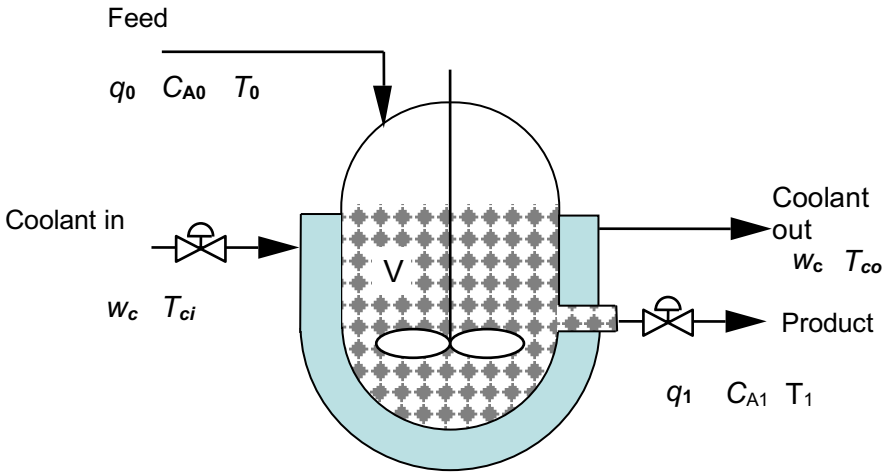
A steady state representation of the distillation column, condenser, reboiler, reflux drum and pumps is used in the design of the equipment, but this cannot be used to inform the control system design. In order to design the control system, the dynamics of the process must be understood.

1.6 Variables

The *controlled* variable is the process variable being regulated by the system. In a feedback controller the controlled variable is usually the *measured* variable. Additional measured variables that are not necessarily controlled may be selected for monitoring the process. In order to achieve the control objective the controlled variable must be matched with one or more variables that can be adjusted, the *manipulated* variables. For example, the temperature controller on the reboiler of a distillation column can adjust the steam flow to regulate the temperature, and hence the composition, at the bottom of the column.

Example 1.4

Reaction A $\rightarrow$ B occurs in a well-mixed CSTR



- Identify the input and output variables and state whether they would normally be measured or unmeasured.
- Decide if input variables are manipulated or disturbances.
- What methods might be employed on the chemical plant to measure these variables.

Variable	Input Manip.	Input Disturb.	Output	Meas	Unmeas	Measurement method
C_{A0}						
q_0						
T_0						
C_{A1}						
q_1						
T_1						
V						
w_c						
T_{ci}						
T_{co}						

1.7 Dynamic Modelling

Process design is generally carried out using steady state models, but in order to design control systems it is necessary to understand process dynamics. This is often achieved through process modelling, using fundamental dynamic heat and material balances, some of which are given on the next page. They will be used in Section 2. These balances are supplemented, as necessary, with rate and constitutive equations e.g.

$$r_A = k_0 e^{-E/RT} C_A$$

$$PV = nRT$$

$$Q = hA\Delta T$$

ACCUMULATION = IN – OUT + CREATED – DESTROYED	
Overall mass, M. w_k is the mass flow rate of the kth stream (positive for flow in, negative for flow out). Component i mass, M_i. w_{ik} is the mass flow rate of component i in the kth stream, r_j is the rate of the jth reaction, ν_{ij} is a stoichiometric coefficient, and m_i is the RMM of component i.	$\frac{dM}{dt} = \sum_{k=1}^{K} w_k$ $\frac{dM_i}{dt} = \sum_{k=1}^{K} w_{ik} + m_i \sum_{j=1}^{J} \nu_{ij} r_j$
CSTR (no rxn). V is volume, q is volumetric flow rate, c is concentration, and subscripts 0 and 1 refer to inlet and outlet conditions.	$\frac{dV}{dt} = q_0 - q_1 \text{ and } \frac{d(Vc_1)}{dt} = q_0 c_0 - q_1 c_1$
PFR (no rxn), delay time t_d.	$C_1(t) = C_0(t - t_d)$, NB velocity may depend on position
CSTR (rxn)	$\frac{dV}{dt} = q_0 - q_1 \text{ and } \frac{d(Vc_1)}{dt} = q_0 c_0 - q_1 c_1 + rV$
Energy (ignoring KE & PE)	$\frac{dU}{dt} = \sum_{k=1}^{K} w_k \hat{H}_k + Q_H + W_s - p\frac{dV}{dt}$.
Energy (liquid, constant M, no rxn)	$MC_p \frac{dT}{dt} = wC_p(T_0 - T_1) + Q_H + W_s$